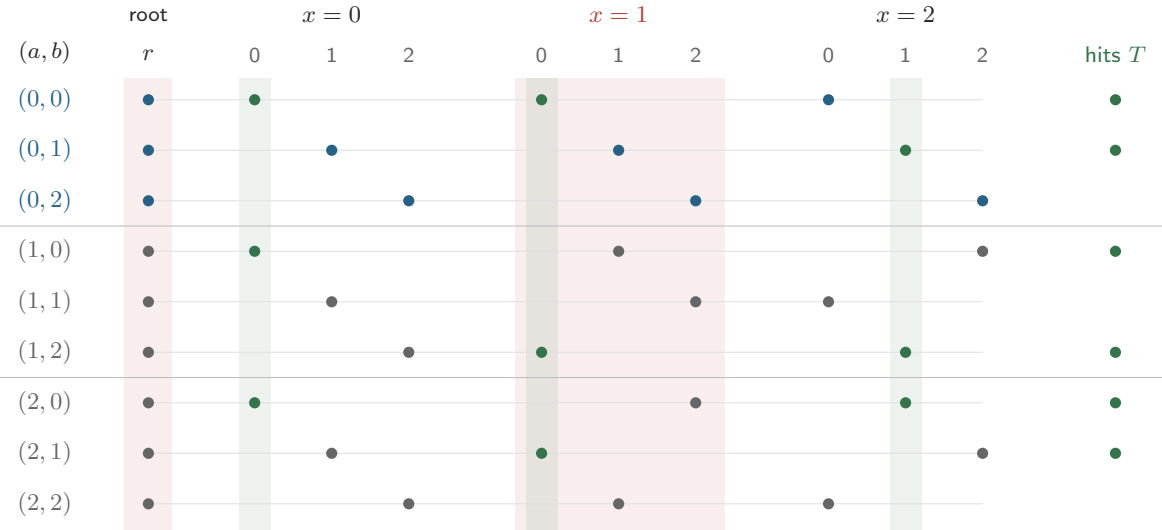


The affine construction at $q = 3$

Nine points $(x,y) \in \mathbb{F}_3^2$, plus a common root r .

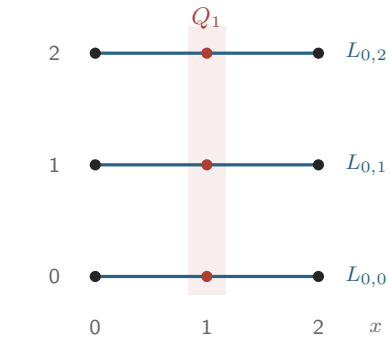
All nine supply edges



Each row is $e_{a,b} = \{r\} \cup \{(x, ax + b) : x \in \mathbb{F}_3\}$.

Green strips mark T ; the six green row markers show supplies hitting T .

A column costs three



Each blue line plus r is a unit-cost supply.

Every supply hits Q_1 once, so $\delta(\{r\} \cup Q_1) = 3$.

The all-cut comparison

For every T : $P(\text{column hit}) \leq \frac{5}{3}P(\text{supply hit})$.

Example: $T = \{(0,0), (1,0), (2,1)\}$.

Columns hit $3/3 = 1$

Supplies hit $6/9 = 2/3$

Ratio: $\frac{1}{2/3} = \frac{3}{2} < \frac{5}{3}$.

Worst cut at $q = 3$: exact maximum $3/2$.

All edges contain $r \notin T$: hitting means crossing.

The finite check does not replace the general proof.